Module 2: Effects of Carcinogen on Rats

Purpose

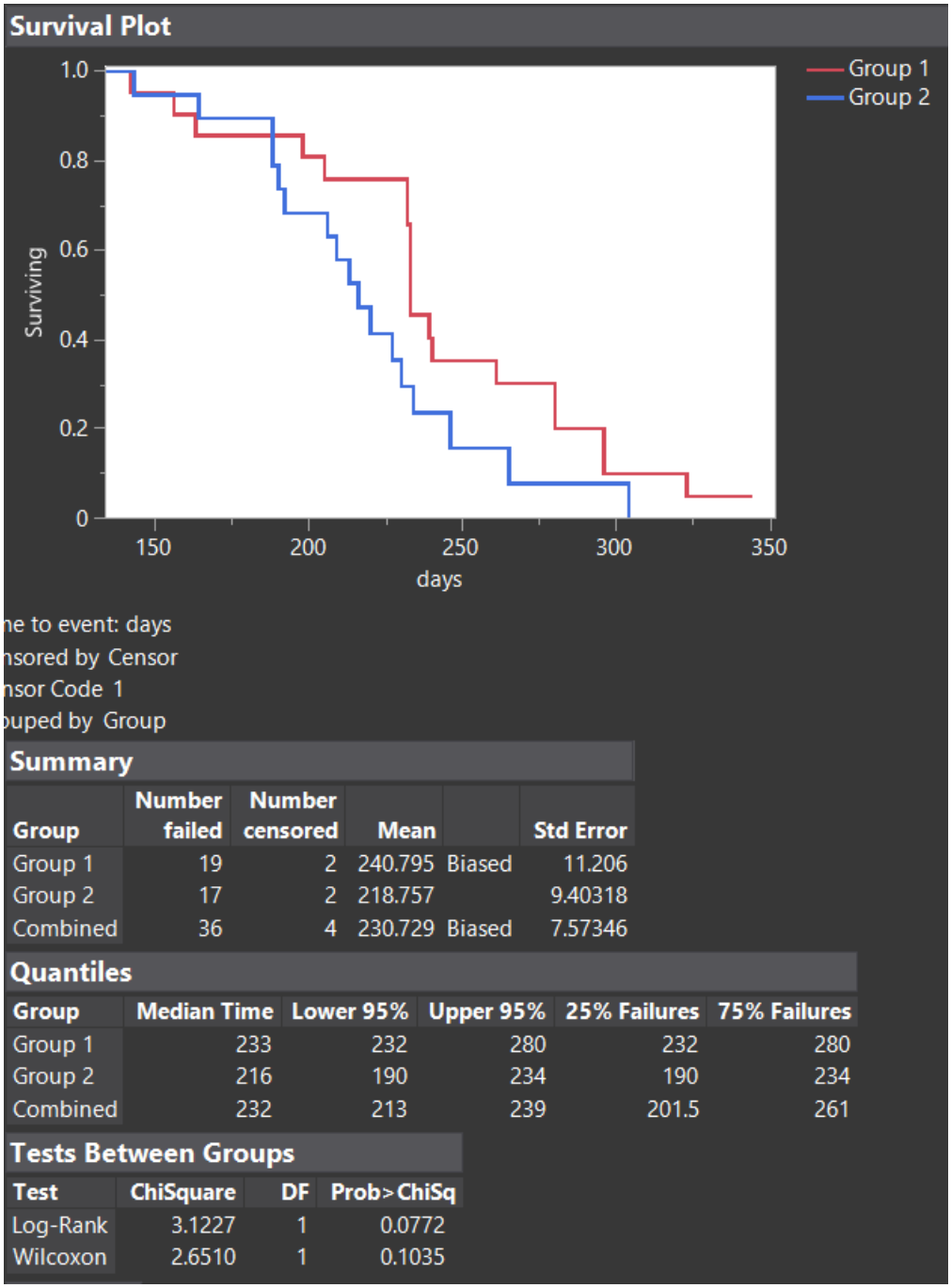
The purpose of this analysis is to compare the survival time between two treatment groups in the dataset to see whether one group lives longer than the other. Kaplan-Meier curves, life-distribution analysis, and Cox proportional hazards model will be used in this analysis.

Kaplan-Meier Curves

The null hypothesis for the Kaplan-Meier comparison is that there will be no difference between the two treatment groups' survivability.

$H_0 : S_1 = S_2$ The survivability of both groups 1 and 2 will be the same.

$H_0 : S_1 \neq S_2$ The survivability of groups 1 and 2 will be different.



ne to event: days
nsored by Censor
nsor Code 1
uped by Group

Summary

Group	Number failed	Number censored	Mean		Std Error
Group 1	19	2	240.795	Biased	11.206
Group 2	17	2	218.757		9.40318
Combined	36	4	230.729	Biased	7.57346

Quantiles

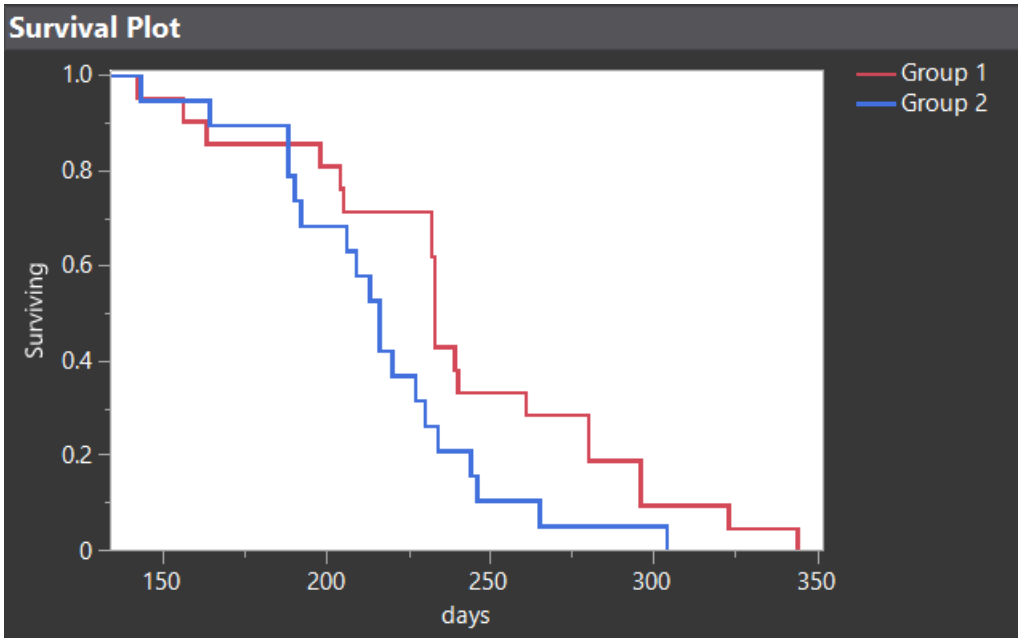
Group	Median Time	Lower 95%	Upper 95%	25% Failures	75% Failures
Group 1	233	232	280	232	280
Group 2	216	190	234	190	234
Combined	232	213	239	201.5	261

Tests Between Groups

Test	ChiSquare	DF	Prob>ChiSq
Log-Rank	3.1227	1	0.0772
Wilcoxon	2.6510	1	0.1035

Above is the Kaplan-Meier curve showing both treatment groups, with censoring, and the tests between groups table. Based on this data the null hypothesis cannot be rejected. Both p-values for Log-Rank (0.077) and Wilcoxon (0.104) tests are higher than the threshold $p<0.05$. The median survival times for each group as shown in the Quantiles Table are 233 days for Group 1 and 216 days for Group 2.

Below is the Kaplan-Meier curve without censoring. It shows that the Log-Rank p-value drops to 0.0485, which is below 0.05. This meets the threshold required to reject the null hypothesis, and shows a statistically significant difference in survival times between groups once censored rats are excluded. In the censored plot rats are included in the



ne to event: days
uped by Group

Summary

Group	Number failed	Number censored	Mean	Std Error
Group 1	21	0	239.19	11.423
Group 2	19	0	215.526	8.28808
Combined	40	0	227.95	7.33283

Quantiles

Group	Median Time	Lower 95%	Upper 95%	25% Failures	75% Failures
Group 1	233	205	261	205	280
Group 2	216	190	230	190	234
Combined	231	209	233	201	245

Tests Between Groups

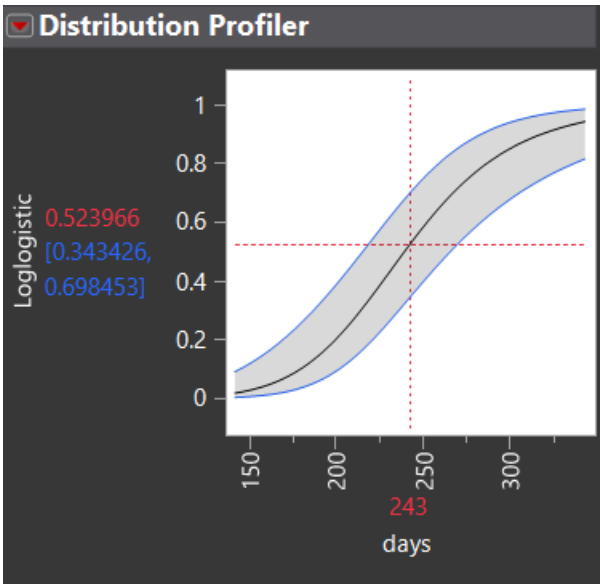
Test	ChiSquare	DF	Prob>ChiSq
Log-Rank	3.8912	1	0.0485*
Wilcoxon	2.8606	1	0.0908

dataset up until the time they are censored. This is why the surviving value plateaus at the end of the curve and does not reach 0. In the non censored plot each rat is treated as if they've experienced death, which underestimates survival and can exaggerate the differences between the groups. This could explain the varying p-values between the groups.

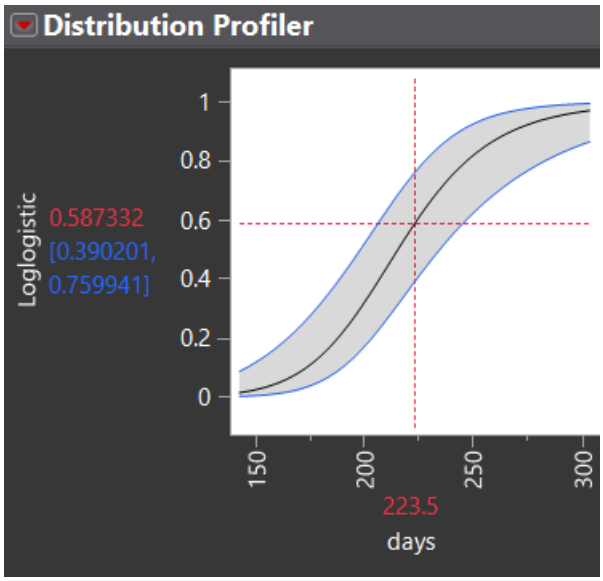
Life Distribution Analysis

The Distribution Profilers below show Group 1 has a mean survival of 243 days and Group 2 has a mean survival of 223.5 days. Group 1 consistently survives longer than Group 2. The Profilers' grey shaded confidence regions visually show that Group 2 experiences deaths over a narrower, steeper time range, but Group 1's survival times are more spread out, with deaths occuring more gradually over the observation period.

Group 1:



Group 2:



Cox Proportional Hazards Model

Below is the Cox Proportional Hazards output. The hazard ratio comparing Group 2 to Group 1 is 1.815, which means that Group 2 rats have about 1.8 times the hazard of death compared to Group 1. The hazard ratio comparing Group 1 to Group 2 is 0.551 which is the reciprocal of Group 2 compared to Group 1, and also indicates that Group 1 has less hazard of death. It is noteworthy, however, that p=0.087, which is above the 0.05 significance level, so the difference is not statistically significant.

Hazard Ratios for Group					
				95% Confidence Interval (Wald)	
Level1	/Level2	Hazard Ratio	Prob>Chisq	Lower	Upper
Group 2	Group 1	1.814656	0.0872	0.91671	3.592166
Group 1	Group 2	0.551069	0.0872	0.278384	1.090857

Summary

Across all analyses, Group 1 showed longer survival than Group 2. The Kaplan-Meier curves showed median survival times of 233 days for Group 1 and 216 days for Group 2. Though the non-censored analysis suggested a significant difference (p=0.0485), the Log-Rank test with censoring failed to reject the null hypothesis. Life Distribution analysis also confirmed that Group 1 survival times are longer and more variable than Group 2 survival times, and Cox Proportional Hazards model estimated a hazard ratio of 1.815 for Group 2 relative to Group 1, but with p = 0.087, which is not statistically significant. Overall, the data suggest a trend with higher survival in Group 1, but the evidence is not strong enough to be statistically sound.